DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

Neural Network Lab - Assessment Report

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Student Information

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Overall Result

25/28 (89.3%) Grade: B

Score by Section

Section	Score	Pct
Activation Functions	5/5	100.0%
Weight Initialization	3/3	100.0%
Loss Functions	3/3	100.0%
Optimizers	2/4	50.0%
Backpropagation	3/3	100.0%
Regularization	3/4	75.0%
Batch Normalization	2/2	100.0%
Architecture Design	4/4	100.0%

Questions to Review

Q13. [Optimizers]

Adam optimizer combines which two techniques?

*Adam maintains m (first moment, like momentum) and v (second moment, like RMSProp). Update: theta -= lr * m_hat / (sqrt(v_hat) + eps).*

Q14. [Optimizers]

Which optimizer typically converges fastest on small datasets?

Adam adapts the learning rate per-parameter using gradient history, making it robust to different gradient scales.

Q20. [Regularization]

How does Dropout prevent overfitting?

With dropout rate=0.5, each neuron is randomly off 50% of training time. This prevents co-adaptation. At inference time, dropout is OFF.